

ADESOLA KAREEM

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PROFESSIONAL SUMMARY

Backend Software Engineer with 2+ years of production Python experience, specializing in distributed pipelines, high-throughput systems, and API design. At WebMD, owned the full Athena vector ingestion pipeline from design to production. Strong in system design, API architecture, and engineering for production scale.

SKILLS

Languages: Python (5+ yrs), JavaScript/Node.js, Java

Backend: FastAPI, REST APIs, Pydantic, JSON Schema, Microservices, Event-Driven Architecture

Data: PostgreSQL, pgvector, Redis, MongoDB, Solr, SQL

AI/LLM: LLM routing, Embeddings (OpenAI, Azure, Anthropic, Vertex, Cohere), RAG, FAISS, Presidio, spaCy

Infra: RabbitMQ, Docker, Kubernetes, GCP, GitLab CI/CD, JWT, OAuth2

Observability: Grafana, Prometheus, Kibana, Structured Logging

WORK EXPERIENCE

WebMD - Software Engineer, Backend

Mar 2025 - Present | Python, FastAPI, RabbitMQ, pgvector, PostgreSQL, Redis, Docker, GCP

- Resolved 6 concurrent performance bottlenecks under a 2M-document load test: async locking, TCP reuse, GC safety, bulk CTE deletes, connection cleanup, and logger crash fix
- Reduced database round-trips to 3 for any batch size via bulk unnest() inserts (delete + doc + chunk/embedding in one transaction)
- Eliminated pod restarts under load by integrating tenacity retry logic (3 attempts, exponential backoff 1s-8s) for transient 502/503/504 errors
- Built pgvector storage backend from scratch: asyncpg, HNSW index, schema-per-business-unit multi-tenant routing, and GIN full-text index
- Designed RabbitMQ batch pipeline: in-memory buffer with configurable flush on BATCH_SIZE or interval; idempotent dedup across 6 composite fields
- Designed multi-backend storage routing (pgvector + Solr) by business unit via env config, with per-backend failure isolation
- Implemented retrieval_mode field (chunk/segment/document) stored across pgvector and Solr, controlling how content surfaces at query time

Nomba - Software Engineer

Nov 2023 - Feb 2024 | Python, MongoDB, Redis, GCP

- Reduced GCP infrastructure costs by 25% through caching strategy and service-level optimization
- Built a high-volume CSV export pipeline backed by MongoDB aggregation and Redis caching

Precious Cornerstone University (NYSC) - Research Assistant

Mar 2024 - Jan 2025

- Trained 20+ faculty on AI tools (ChatGPT, Claude), improving team productivity by ~40%

EDUCATION

Obafemi Awolowo University, Ile-Ife, Nigeria | Aug 2023

BSc Computer Science with Mathematics - First Class Honours (GPA: 4.62)

Scholarships: NNPC-CHEVRON National University Scholarship, Adewale Obadimu Scholarship

CERTIFICATIONS

HackerRank: Java (2024), REST API (2023), Problem Solving (2023), Python (2022)

PROJECTS

Scalable Notification Service | May 2026 - Present

FastAPI, RabbitMQ, Redis, PostgreSQL, Docker, Prometheus, Grafana | github.com/kareemadesola/scalable-notification-service

- Production-grade multi-channel notification system (email, SMS, push, WebSocket) with independent channel scaling and dead-letter queues
- Built Redis rate limiting and Prometheus + Grafana observability stack; designed for ~17K notifications/sec throughput

AI Investigation and Research Assistant | Apr 2026

Python, FAISS, RAG, Agentic pipeline | github.com/kareemadesola/ai-investigation-assistant

- Agentic RAG system combining FAISS vector search, multi-step reasoning, and a hallucination detection evaluation layer